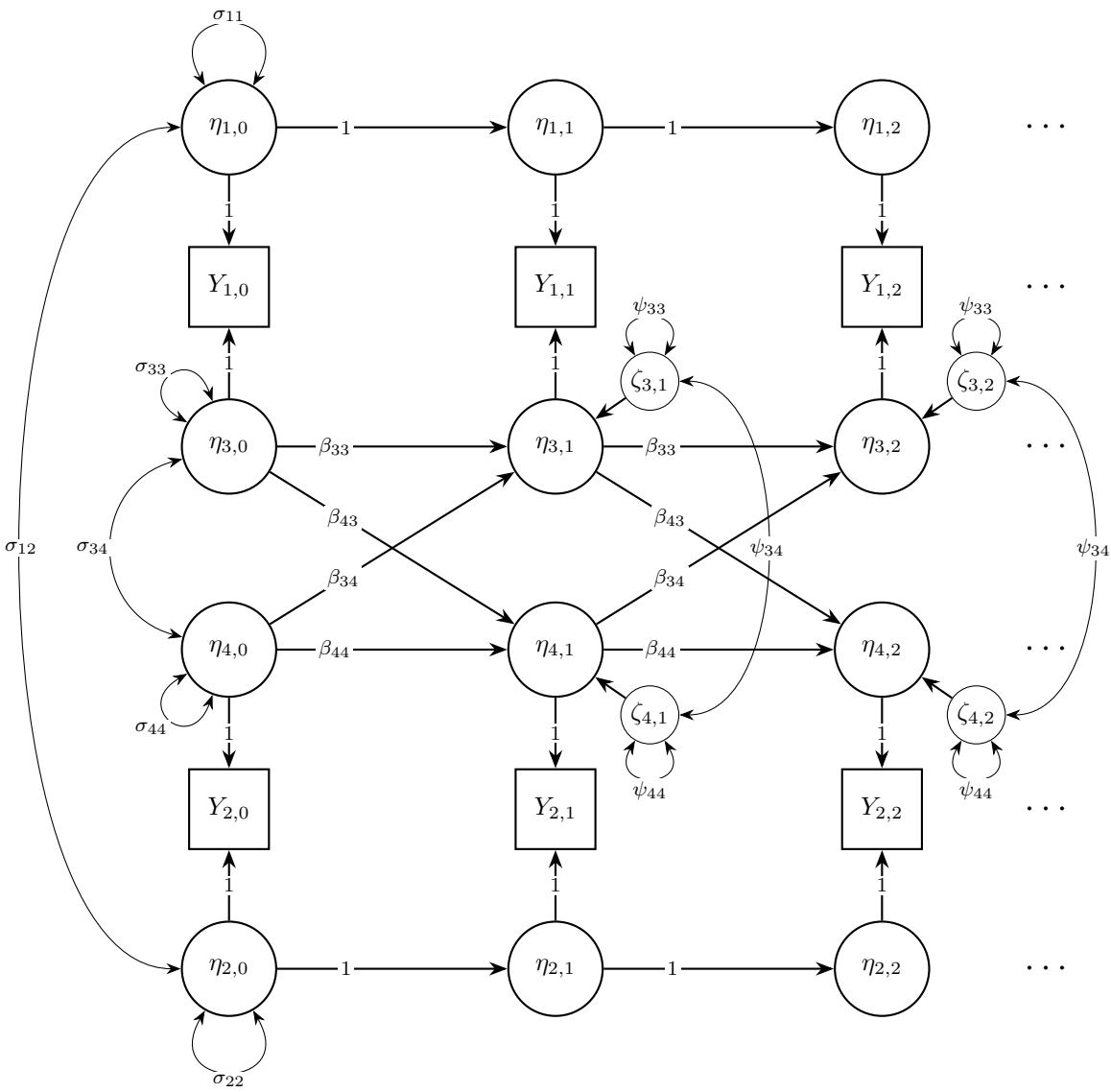



$\boldsymbol{\eta}_t = (\eta_{1,t}, \eta_{2,t}, \eta_{3,t}, \eta_{4,t})'$, $\quad \mathbf{y}_t = (Y_{1,t}, Y_{2,t})'$; no measurement error

traits $\eta_{1,t} = \eta_1$ and $\eta_{2,t} = \eta_2$ for all $t$; within-person components $\eta_{3,t}$ and $\eta_{4,t}$

$$\boldsymbol{\beta} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \beta_{33} & \beta_{34} \\ 0 & 0 & \beta_{43} & \beta_{44} \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{\Psi} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \psi_{33} & \psi_{34} \\ 0 & 0 & \psi_{43} & \psi_{44} \end{pmatrix}$$

$$\boldsymbol{\eta}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\mu}_0 = \begin{pmatrix} \mu_1 \\ \mu_2 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} \sigma_{11} & \sigma_{12} & 0 & 0 \\ \sigma_{21} & \sigma_{22} & 0 & 0 \\ 0 & 0 & \sigma_{33} & \sigma_{34} \\ 0 & 0 & \sigma_{43} & \sigma_{44} \end{pmatrix}$$